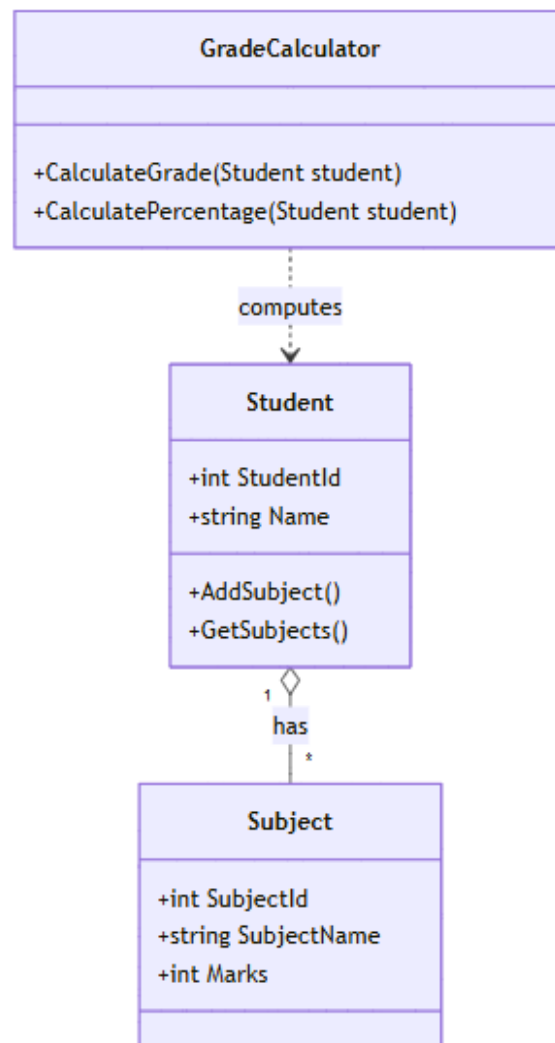


Class, Object and Sequence Diagram

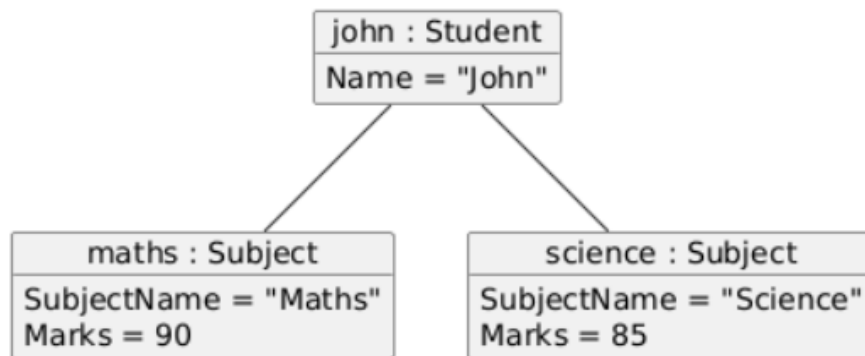
Problem 1: School Result Application

- **Class Diagram**
 - Diagram Description:
 - Classes: Student, Subject, GradeCalculator
 - Relationships:
 - A Student has multiple Subject entries (Aggregation).
 - GradeCalculator computes the results for a Student.



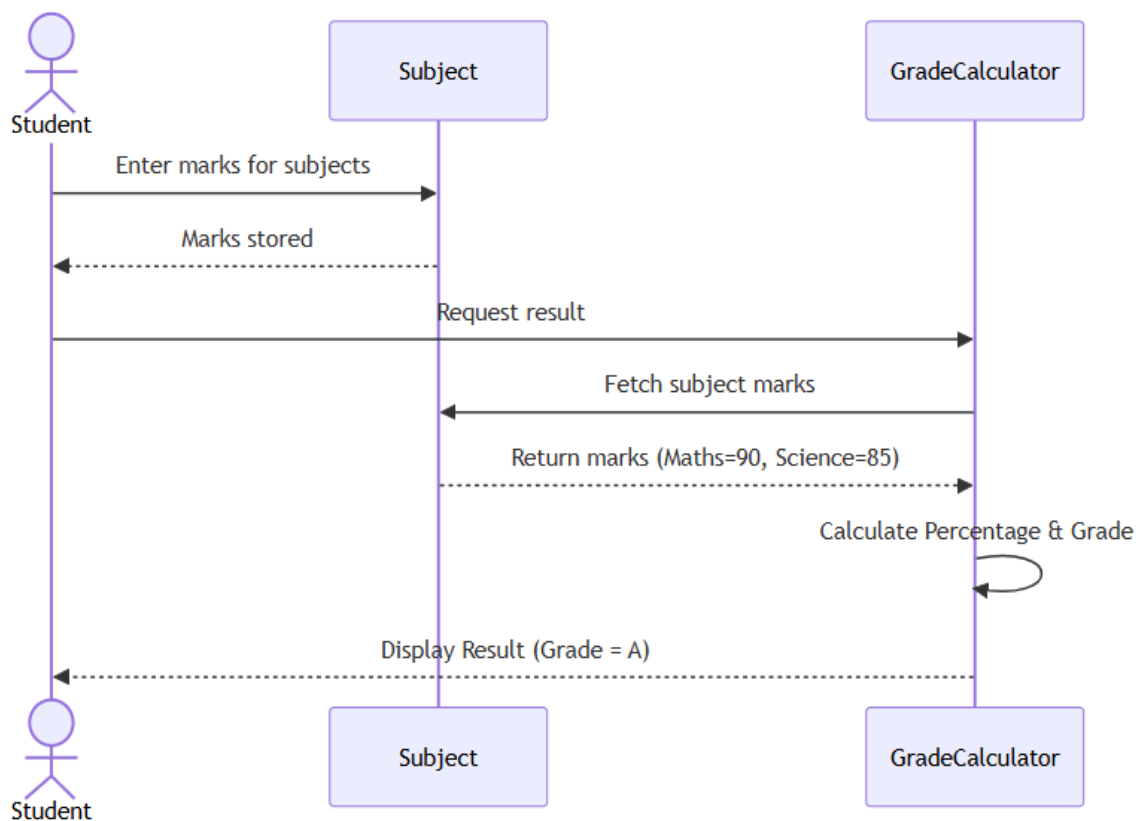
- **Object Diagram**

- Example:
 - Student: John
 - Subjects: Maths, Science
 - Marks: 90, 85



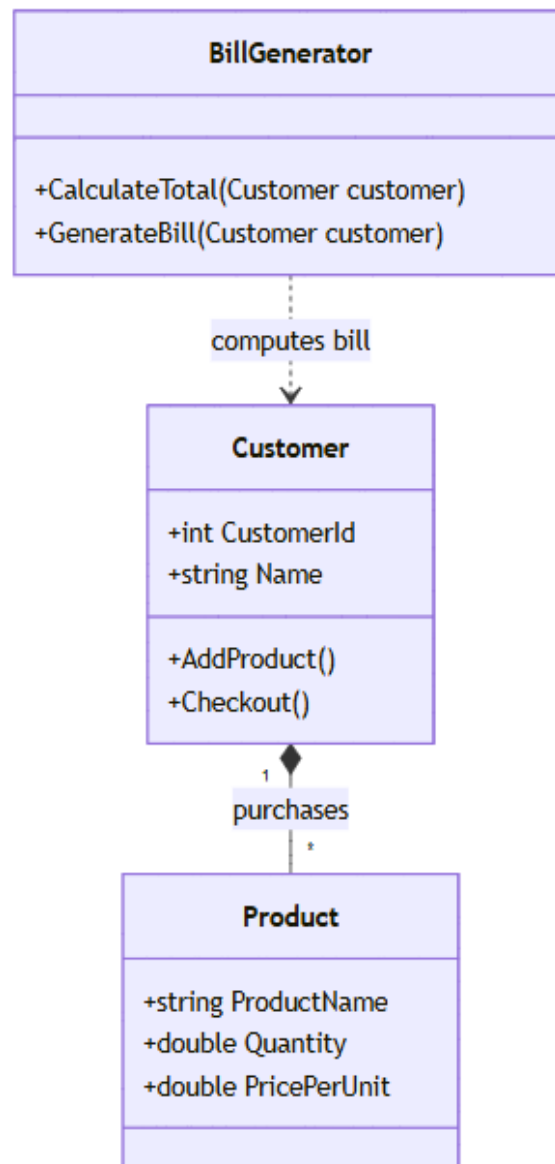
- **Sequence Diagram**

- Scenario: A student requests their grade based on marks in subjects.
- Actors: Student, GradeCalculator



Problem 2: Grocery Store Bill Generation Application

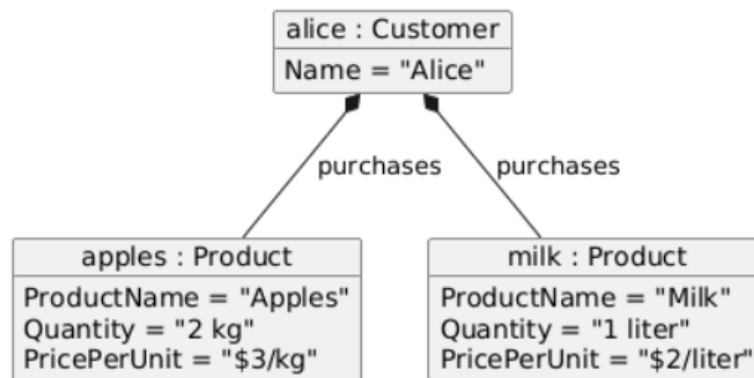
- **Class Diagram**
 - Diagram Description:
 - Classes: Customer, Product, BillGenerator
 - Relationships:
 - A Customer can purchase multiple Product items (Composition).



- **Object Diagram**

- **Example:**

- Customer: Alice
 - Products:
 - Apples (2 kg at \$3 per kg)
 - Milk (1 liter at \$2 per liter)



- **Sequence Diagram**

- **Scenario:** A customer checks out at the grocery store, and the total bill is generated.
 - **Actors:** Customer, BillGenerator

